

BICC1 Interacts with PKD1 and PKD2 to Drive Cystogenesis in ADPKD

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eLife Assessment

This study presented **valuable** findings regarding the basic molecular pathways leading to the cystogenesis of Autosomal Dominant Polycystic Kidney Disease, suggesting BICC1 functions as both a minor causative gene for PKD and a modifier of PKD severity. Although some **solid** data were supplied to show the functional and structural interactions between BICC-1 and PKD2 and their relevance to the pathogenesis of ADPKD, the characterization of such interactions appear to be **incomplete**, which renders the specific relevance of these findings for disease etiology unclear.

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Abstract

Autosomal dominant polycystic kidney disease (ADPKD) is primarily of adult-onset and caused by pathogenic variants in *PKD1* or *PKD2*. Yet, disease expression is highly variable and includes very early-onset PKD presentations *in utero* or infancy. In animal models, the RNA-binding molecule Bicc1 has been shown to play a crucial role in the pathogenesis of PKD. To study the interaction between *BICC1*, *PKD1* and *PKD2* we combined biochemical approaches, knockout studies in mice and *Xenopus*, genetic engineered human kidney cells as well as genetic association studies in a large ADPKD cohort. We first demonstrated that BICC1 physically binds to the proteins Polycystin-1 and -2 encoded by *PKD1* and *PKD2* via distinct protein domains. Furthermore, PKD was aggravated in loss-of-function studies in *Xenopus* and mouse models resulting in more severe disease when *Bicc1* was depleted in conjunction with *Pkd1* or *Pkd2*. Finally, in a large human patient cohort, we identified a sibling pair with a homozygous *BICC1* variant and patients with very early onset PKD (VEO-PKD) that exhibited compound heterozygosity of *BICC1* in conjunction with *PKD1* and *PKD2* variants. Genome editing demonstrated that these *BICC1* variants were hypomorphic in nature and impacted

disease-relevant signaling pathways. These findings support the hypothesis that *BICC1* cooperates functionally with *PKD1* and *PKD2*, and that *BICC1* variants may aggravate PKD severity highlighting RNA metabolism as an important new concept for disease modification in ADPKD.

Introduction

Autosomal dominant polycystic kidney disease (ADPKD) is the most frequent life-threatening genetic disease and one of the most common Mendelian human disorders with an estimated prevalence of 1/400-1000.^{1,2} This equates to around 12.5 million affected individuals worldwide. About 5-10% of all patients requiring renal replacement therapy are affected by ADPKD. The majority of ADPKD patients carry a pathogenic germline variant in the *PKD1* or *PKD2* gene and present with the disease in adulthood.³⁻⁴ However, occasionally, ADPKD can manifest in infancy or early childhood [< 2 years, very-early onset ADPKD (VEO-ADPKD)], and in late childhood or early teenage years [2-16 years, early-onset ADPKD (EO-ADPKD)].^{5,6} VEO patients and fetuses often present with a Potter sequence and significant peri- or neonatal demise, which can be clinically indistinguishable from a typical Autosomal Recessive Polycystic Kidney Disease (ARPKD) presentation caused by *PKHD1* mutations.^{7,8} However, in contrast to VEO/EO-ADPKD, ARPKD kidneys invariably manifest as fusiform dilations of renal collecting ducts and distal tubules that usually remain in contact with the urinary system.⁴ Co-inheritance of an inactivating *PKD1* or *PKD2* mutation with an incompletely penetrant minor PKD allele *in trans* provides a likely explanation for VEO-ADPKD.⁹ In fact, we recently reported that the majority (70%) of VEO-ADPKD cases in an international diagnostic cohort had biallelic *PKD1* variants (i.e., a pathogenic variant *in trans* with a hypomorphic, low penetrance variant), while cases of biallelic *PKD2* and digenic *PKD1/PKD2* were rather rare.¹⁰ In line with the dosage theory for PKD,¹¹ several other genes (*GANAB*, *DNAJB11*, *ALG8*, *ALG9*, *IFT140*) have been associated with a dominant, but late-onset atypical adult presentation and sometimes incomplete penetrance.^{4,12-15} Yet, not all VEO/EO-ADPKD patients can be explained by monogenic inheritance suggesting digenic or oligogenic inheritance causes.

Previous data from mouse, *Xenopus* and zebrafish suggest a crucial role for the RNA-binding protein Bicc1 in the pathogenesis of PKD, although *BICC1* mutations in human PKD have not been previously reported.¹⁶⁻²² *BICC1* encodes an evolutionarily conserved protein that is characterized by 3 K-homology (KH) and 2 KH-like (KHL) RNA-binding domains at the N-terminus and a SAM domain at the C-terminus, which are separated by a by a disordered intervening sequence (IVS).²³⁻²⁸ The protein localizes to cytoplasmic foci involved in RNA metabolism and has been shown to regulate the expression of several genes such as *Pkd2*, *Adcyd6* and *Pkia* in the kidney.^{22,29} We now present data providing a mechanistic model linking *BICC1* with the three major cystic proteins. We show that *BICC1* physically interacts with the *PKD1* (PC1) and the *PKD2* (PC2) protein in human kidney cells. We also demonstrate that *Pkd1* and *Pkd2* modifies the cystic phenotype in *Bicc1* mice in a dose-dependent manner and that Bicc1 functionally interacts with *Pkd1*, *Pkd2* and *Pkhd1* in the pronephros of *Xenopus* embryos. Finally, this interaction is supported by human patient data where *Bicc1* alone or in conjunction with *PKD1* or *PKD2* is involved in VEO-PKD.

Methods

Cell Culture and Biochemical Studies

The characterization of the interaction between BICC1, PC1 and PC2 as well as the analysis of the human BICC1 variants were performed using standard approaches detailed in the Supplementary Methods.

Animal Studies

Mouse and *Xenopus laevis* studies were approved by the Institutional Animal Care and Use Committee at the Cleveland Clinic Foundation and LSU Health Sciences Center (present and former employer of Dr. Wessely) and adhered to the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Experimental design and data interpretation followed the ARRIVE1 reporting guidelines.³⁰

International Diagnostic Clinical Cohort

Research was performed following written informed consent and according to the declaration of Helsinki and oversight was provided by the Medizinische Genetik Mainz. DNA extraction was performed according to standard procedures (see Supplementary Methods for details).

Statistical analysis

Data are presented as mean values $\pm$ SEM. Paired and unpaired two-sided Student's t test or ANOVA were used for statistical analyses with a minimum of $P < 0.05$ indicating statistical significance. Measurements were taken from distinct biological samples. Analyses were carried out using Prism 10 (Graphpad).

Results

Interaction of BICC1 with PC1 and PC2

Loss of Pkd1 has been associated with lower expression of Bicc1.³¹ Furthermore, Bicc1 has been shown to regulate *Pkd2* expression in cellular and animal models.^{22, 32, 33} However, whether this is due to direct protein-protein interactions between BICC1, PKD1 (PC1) and PKD2 protein (PC2) have not been investigated. In pilot experiments, BICC1 was detected by mass spectrometry in a pulldown assay from cells stably expressing a Polycystin-1 PLAT domain (Polycystin-1, Lipoxigenase, Alpha-Toxin)-YFP fusion.³⁴ The direct binding between the PC1-PLAT domain and Bicc1 was confirmed using *in vitro* binding assays, but also detected binding to the PC1 C-terminus (CT1) (Supplementary Fig. S1a,c).

Utilizing recombinant GST domains of PC1 and PC2, we demonstrated that BICC1 binds to both proteins in GST-pulldown assays (Fig. 1a, b). In the case of PC1, myc-mBICC1 strongly interacted with its C-terminus (GST-CT1) but its interaction was abolished by a PC1-R4227X truncation mutation (GST-CT1-R4227X) (Fig. 1b,c). In the case of PC2, myc-mBicc1 associated with both recombinant GST N-terminal (GST-NT2) and C-terminal (GST-CT2) fusions. To investigate whether binding was direct or indirect, we performed *in vitro* binding assays using *in vitro* translated myc-Bicc1 and recombinant PC1 and PC2 domains. GST-pulldowns confirmed a direct interaction between myc-Bicc1 and GST-CT1 but not GST-CT1-R4227X (Fig. 1d, e). Similarly, myc-Bicc1 interacted directly with GST-NT2. While binding was stronger with the distal sequence (NT2 aa101-223) both N-terminal fragments contributed to the overall binding to Bicc1 (Fig. 1d, e). Interestingly, no direct interaction between Bicc1 and GST-CT2 was detected (Supplementary Fig. S1b) suggesting that the observed *in vivo* interaction with Bicc1 is indirect. Finally,

immunoprecipitation using lysates from human kidney epithelial cells (UCL93) to assay endogenous, non-overexpressed proteins showed that PC1, PC2 and BICC1 form protein complexes *in vivo* (Fig. 1f, g).

Different Interaction Motifs for the Binding of Bicc1 to the Polycystins

To define the PC1/PC2 interaction domain(s) in Bicc1, we generated deletion constructs lacking the SAM domain (myc-mBicc1-ΔSAM) or the KH/KHL domains (myc-mBicc1-ΔKH) (Fig. 2a) and studied them by co-IP. Full-length PC1 co-immunoprecipitated with full-length myc-mBicc1 (Fig. 2b,c). Deleting the SAM domain reduced the association to PC1 by ~55% compared to full-length myc-mBicc1. Surprisingly, an 8-fold stronger interaction was observed between full-length PC1 and myc-mBicc1-ΔKH compared to myc-mBicc1 or myc-mBicc1-ΔSAM. These results suggested that the interaction between PC1 and Bicc1 involves the SAM but not the KH/KHL domains (or the first 132 amino acids of Bicc1). It also suggests that the N-terminus could have an inhibitory effect on PC1-BICC1 association.

Similar experiments were performed to define the Bicc1 interacting domains for PC2 (Fig. 2d,e). Full-length PC2 (PC2-HA) interacted with full-length myc-mBICC1. Unlike PC1, PC2 interacted with myc-mBICC1-ΔSAM, but not myc-mBICC1-ΔKH suggesting that PC2 binding is dependent on the N-terminal domains but not the SAM domain. Co-expression of BICC1, deletion constructs lacking the SAM domain (myc-mBicc1-ΔSAM) or the KH domains (myc-mBicc1-ΔKH) however had no effect on the interaction of PC1 with PC2 in co-immunoprecipitation assays (Fig 2f,g) suggesting that these interactions are not mutually exclusive.

Cooperativity of BICC1 with other PKD genes

Since our biochemical analysis indicated a direct interaction between BICC1, PC1 and PC2, we wondered whether this is biologically relevant. If this were the case, BICC1 should cooperate with other PKD genes and reducing BICC1 activity in conjunction with reducing either PKD1 or PKD2 activity should still cause a cystic phenotype. We first addressed this question in the *Xenopus* system (Fig. 3), which is an easily manipulatable model to study PKD. The PKD phenotype in frog is characterized by dilated kidney tubules, the loss of the expression of the sodium bicarbonate cotransporter 1 (Nbc1) in the distal tubule and the emergence of body-wide edema as a sign of a malfunctioning kidney.^{21, 22, 35, 36} Knockdown of Bicc1, Pkd1, Pkd2 or the ARPKD gene *Pkhd1* caused a PKD phenotype (Fig. 3e-i and Supplementary Fig. S2a). The latter, *Pkhd1* was included to assay not only ADPKD, but also ARPKD, which is generally thought to disturb the same cellular mechanisms. To test whether Bicc1 cooperated with the PKD genes we then performed combined knockdowns. We titrated each of the four MOs to a concentration that on its own resulted in little phenotypic changes upon injection into *Xenopus* embryos (Fig. 3j,k and Supplementary Fig. S2b). However, combining *Bicc1*-MO1+2 with *Pkd1*-sMO, *Pkd2*-MO or *Pkhd1*-sMO at suboptimal concentrations resulted in the re-emergence of a strong PKD phenotype.

While injections with individual MOs developed edema in about 10% of the embryos, co-injections caused edema formation in almost 100% of the embryos (Fig. 3j, last 3 columns). A similar result was seen for the expression of *Nbc1* in the late distal tubule, where individual MO injections showed some changes in gene expression, but double MO injections had a highly synergistic effect resulting in a near complete loss of *Nbc1* (Fig. 3k).

We next investigated whether a similar cooperation between Bicc1 and Pkd1/Pkd2 can be observed in a genetic model. We initially focused on Bicc1 and Pkd2. Both *Bicc1* and *Pkd2* knockout mice develop cystic kidneys as early as E15.5.^{22, 37} As this is the earliest time point cystic kidneys can be observed, crossing those strains did not allow us to assess cooperativity (data not

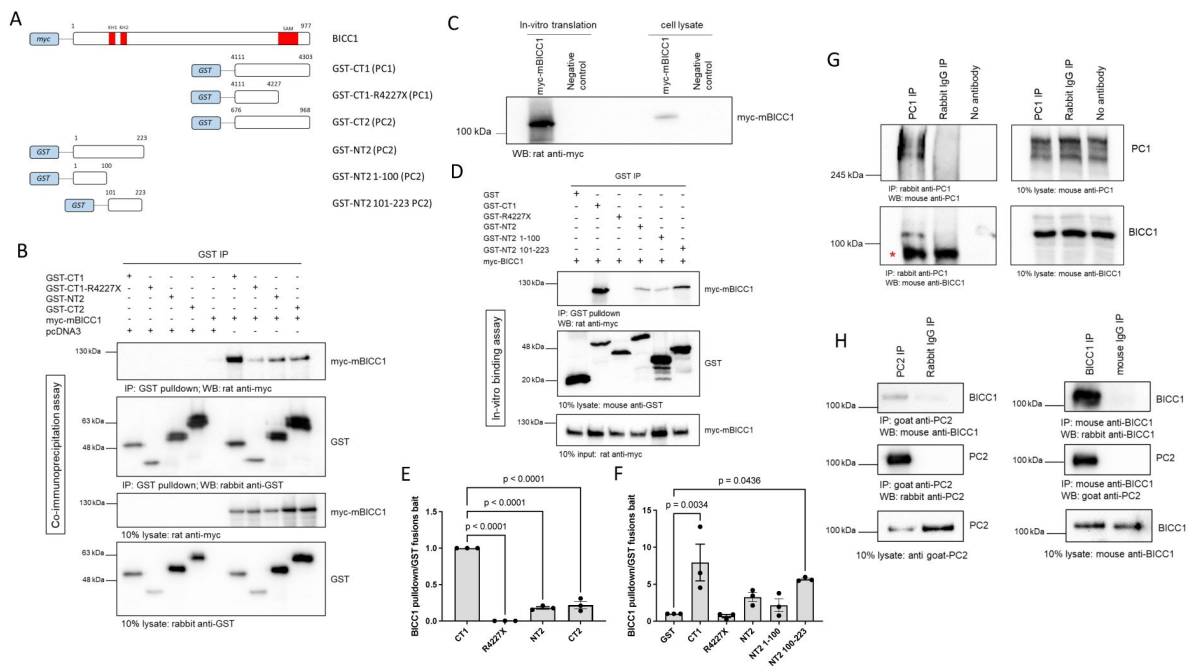


Figure 1.

***Bicc1* Forms a Complex with Polycystin-1 and Polycystin-2.**

Full-length and deletion myc-tagged constructs of *Bicc1* were co-expressed with either full-length HA-tagged PC1 or PC2 in HEK-293 cells and tested for their ability to interact by co-IP. (a) Schematic diagram of the constructs used in this experiment. (b) Western blot analysis following co-IP experiments, using GST tagged constructs as bait, identified protein interactions between PC1 or PC2 domains and *Bicc1*. pcDNA3 was included as a negative control. CT = C-terminus, NT = N-terminus, GST = Glutathione S-Transferase. Co-IP experiments (n=3) were quantified in (c). (d) Western blot analysis following *in-vitro* pulldown experiments, using purified GST tagged constructs as bait, identified direct protein interactions between PC1 or PC2 domains and *in vitro* translated myc-Bicc1. In-vitro binding experiments (n=3) were quantified in (e). (f) Western blot analysis following co-IP experiments, using a rabbit PC1 antibody (2b7) as bait, identified protein interactions between endogenous PC1 and BICC1 in UCL93 cells. A non-immune rabbit IgG antibody or no antibody was included as a negative control; * denotes a non-specific IgG band. (g) Western blot analysis following co-IP experiments, using an anti-mouse Bicc1 or anti-goat PC2 antibody as bait, identified protein interactions between endogenous PC2 and BICC1 in UCL93 cells. Non-immune goat and mouse IgG was included as a negative control.

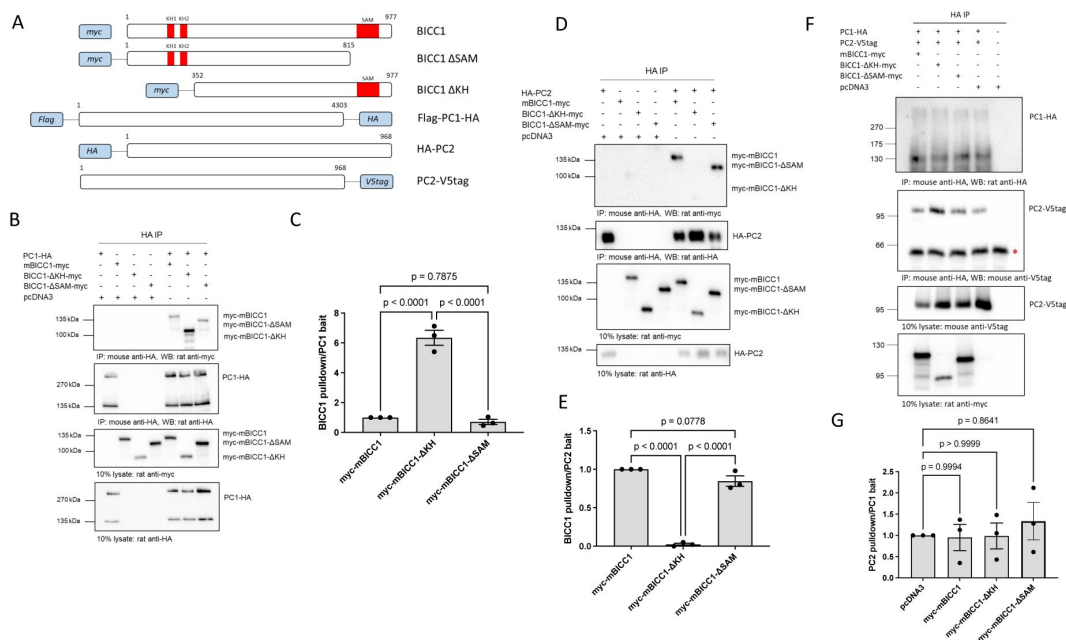


Figure 2.

Interactions between *Bicc1* and Polycystin1/2 Require Different Binding Motifs.

Full-length and deletion myc-tagged constructs of *Bicc1* were co-expressed with either full-length HA-tagged PC1 or PC2 in HEK-293 cells and tested for their ability to interact by co-IP. **(a)** Schematic diagram of the constructs used in this set of experiments with the amino acid positions of full-length *Bicc1* or the different deletions indicated. **(b,c)** Western blot analysis following co-IP experiments, using a PC1-HA-tagged construct as bait, identified protein interactions between PC1 and *Bicc1* domains. pcDNA3 was included as a negative control **(b)**. co-IP experiments ($n=3$) were quantified in **(c)**. **(d,e)** Western blot analysis following co-IP experiments, using a PC2-HA tagged construct as bait, identified protein interactions between PC2 and *Bicc1* domains **(d)**. pcDNA3 was included as a negative control. Quantification of the co-IP experiments ($n=3$) is shown in **(e)**. **(f, g)** Western blot analysis following co-IP experiments, using a PC1-HA-tagged construct as bait. The interaction between PC1 and PC2 was not altered in the presence of either full-length *Bicc1* or its deletion domains. pcDNA3 was included as a negative control. Asterix represents non-specific interaction with mouse IgG. **(f)**. co-IP experiments ($n=3$) were quantified in **(g)**. One way ANOVA comparisons were performed to assess significance and *P* values are indicated. Error bars represent standard error of the mean.

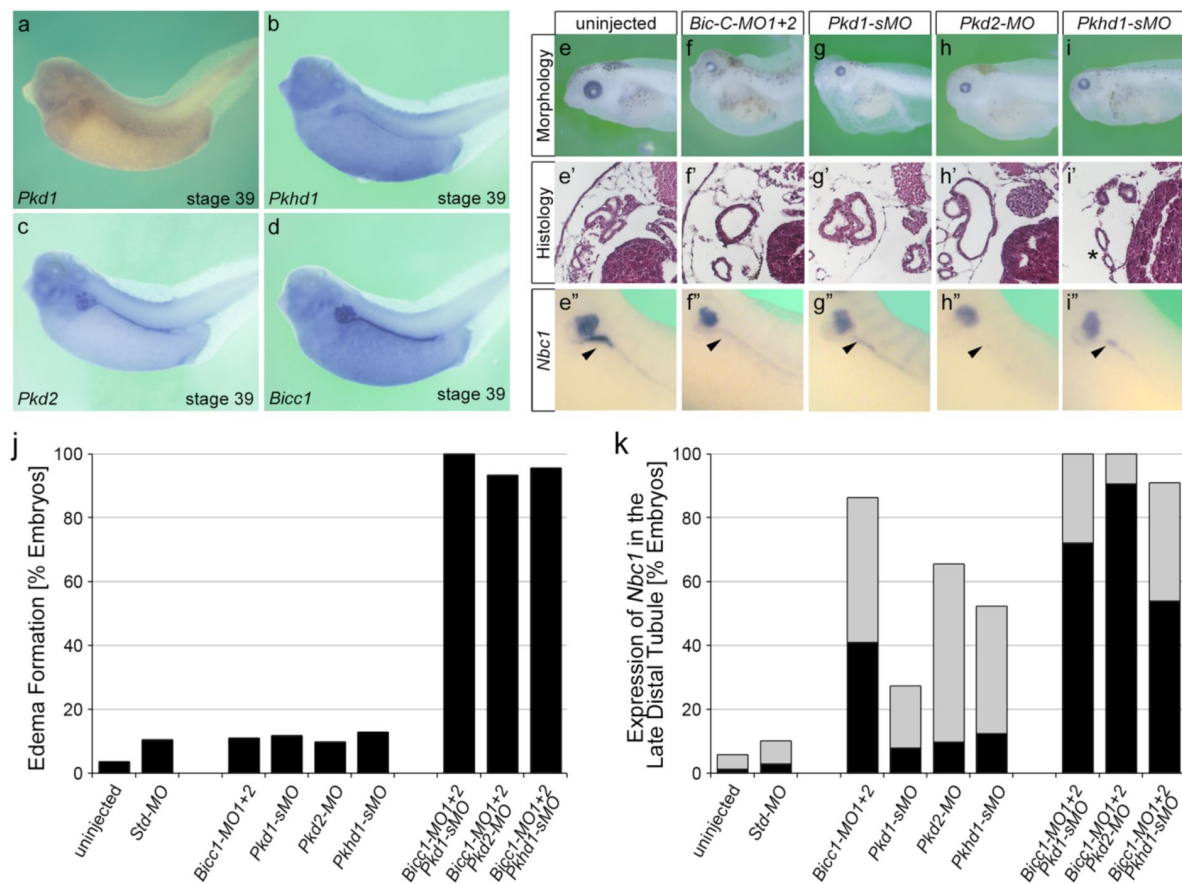


Figure 3.

Cooperativity of Bicc1 and PKD Genes in *Xenopus*.

(a-d) mRNA expression of *Pkd1*, *Pkhd1*, *Pkd2* and *Bicc1* in the *Xenopus* pronephros at stage 39. (e-i'') Knockdown of Bicc1 (f-f''), *Pkd1* (g-g''), *Pkd2* (h-h'') and *Pkhd1* (i-i'') by antisense morpholino oligomers (MOs) results in a PKD phenotype compared to uninjected control *Xenopus* embryos (e-e''). The phenotype is characterized by the formation of edema due to kidney dysfunction (e,f,g,h,i; stage 43), the development of dilated renal tubules (e',f',g',h',i'; stage 43) and the loss of *Nbc1* in the late distal tubule by whole mount *in situ* hybridizations (arrowheads in e'',f'',g'',h'',i''); stage 39). (j,k) To examine cooperativity, *Xenopus* embryos were injected with suboptimal amounts of the MOs, either alone or in combination, and analyzed for edema formation at stage 43 (j) and the expression of *Nbc1* at stage 39 (k). More than three independent experiments were analyzed. In (k), gray bars show reduced and black bars show absent expression of *Nbc1* in the late distal tubule. Data are the accumulation of multiple independent fertilizations with N>30 for each condition.

shown). Moreover, like in the case of compound *Pkd1/Pkd2* mutants,³⁸ kidneys from *Bicc1^{+/-}·Pkd2^{+/-}* did not exhibit cysts (data not shown). Thus, we instead used mice carrying the *Bicc1* hypomorphic allele *Bpk*, which develop a cystic kidney phenotype postnatally.^{18, 39} To assess cooperativity, we removed one copy of *Pkd2* in the *Bpk* mice. Comparing the kidneys of *Bicc1^{Bpk/Bpk}·Pkd2^{+/-}* to those of *Bicc1^{Bpk/Bpk}·Pkd2^{+/+}* at postnatal day P14 revealed that the compound mutant kidneys were larger and more translucent (**Fig. 4a**) and the kidney/body weight ratios (KW/BW) were significantly increased (**Fig. 4b**). Moreover, analyzing survival the compound mutants showed a trend towards an earlier demise (**Supplementary Table S1**). We did not detect sex differences in the phenotype (**Supplementary Figure S3c**). Yet, the reduction in *Pkd2* gene dose affected the progression of the disease, but not its onset. Performing the same analysis at postnatal day P4 did not show any differences (**Fig. 4c**).

Next, we performed a similar mouse study for *Pkd1* by reducing the gene dose of *Pkd1* postnatally in the collecting ducts using a *Pkhd1-Cre* as previously described⁴⁰ (in the following referred to as *Pkd1^{CD}*). Similar to the *Bicc1/Pkd2* scenario, *Bicc1^{Bpk/Bpk}·Pkd1^{+/-CD}* appeared larger when comparing kidneys from littermates (**Fig. 4d**) and littermates exhibited statistically significant differences in KW/BW ratio (**Fig. 4e**). Yet, the phenotype was rather subtle and aggregating all the data did not show differences in KW/BW ratios between *Bicc1^{Bpk/Bpk}·Pkd1^{+/+}* and *Bicc1^{Bpk/Bpk}·Pkd1^{+/-CD}* mice (**Supplementary Fig. S3d**). Thus, to further corroborate the genetic interaction, we determined the cystic index for proximal tubules and collecting ducts using LTA and DBA staining, respectively. This showed an increase in collecting duct cysts upon removal of one copy of *Pkd1* (**Fig. 4g**). Like in the case of *Pkd2*, the phenotype seems to be correlated with cyst expansion and not the onset, as there was no difference at postnatal day P7 (**Fig. 4f**) and we did not detect increased mortality in the compound mutants (**Supplementary Table S2**). It is noteworthy that neither the *Bicc1/Pkd2* nor the *Bicc1/Pkd1* compound mutants showed an aggravated kidney function based on Blood Urea Nitrogen (BUN) levels (**Supplementary Fig. S3a,b,e**) likely due to the aggressive nature of the *Bicc1^{Bpk/Bpk}* phenotype. Together these data supported our biochemical interaction data and demonstrated that *BICC1* cooperated with *PKD1* and *PKD2*.

Finally, to better understand how *Bicc1* would exert such a phenotype, we analyzed the expression of the PKD genes in the *Bicc1^{Bpk/Bpk}* mice. We have previously demonstrated that *Pkd2* levels are reduced in a complete *Bicc1* null mice,²² performing qRT-PCR of P4 kidneys (i.e. before the onset of a strong cystic phenotype), revealed that *Bicc1*, *Pkd1* and *Pkd2* were statistically significantly down-regulated (**Fig. 4h-j**). The effect on *Pkd2* was confirmed by protein analysis for PC2 (**Fig. 4k** and **Supplementary Fig. S3f**).

***BICC1* Variants in Patients with early and severe Polycystic Kidney Disease**

To evaluate whether these interactions are relevant for human PKD, we analyzed an international cohort of 2,914 PKD patients by massive parallel sequencing (MPS)^{41, 42} focusing on VEO-ADPKD patients with the hypothesis that *BICC1* variants may lead to a more severe and earlier PKD phenotype. While variants in *BICC1* are very rare, we could identify two patients with *BICC1* variants harboring an additional *PKD2* or *PKD1* variant *in trans*, respectively. None of these *BICC1* variants were detected in the control cohort or in non-VEO-ADPKD patients. Moreover, besides the variants reported below, the patients had no other variants in any of other PKD genes or genes which phenocopy PKD including *PKD1*, *PKD2*, *PKHD1*, *HNF1B*, *GANAB*, *IFT140*, *DZIP1L*, *CYS1*, *DNAJB11*, *ALG5*, *ALG8*, *ALG9*, *LRP5*, *NEK8*, *OFD1*, or *PMM2*. The first patient was severely and prenatally affected demonstrating a Potter sequence with huge echogenic kidneys and oligo-/anhydramnios. Autopsy confirmed VEO-ADPKD with absence of ductal plate malformation invariably seen in ARPKD. The fetus carried the *BICC1* variant (c.2462G>A, p.Gly821Glu) inherited from his father, who presented with two small renal cysts in one of his kidneys, and a *PKD2* variant (c.1894T>C, p.Cys632Arg) that arose *de novo* (**Fig. 5a**). Individual *in silico* predictions

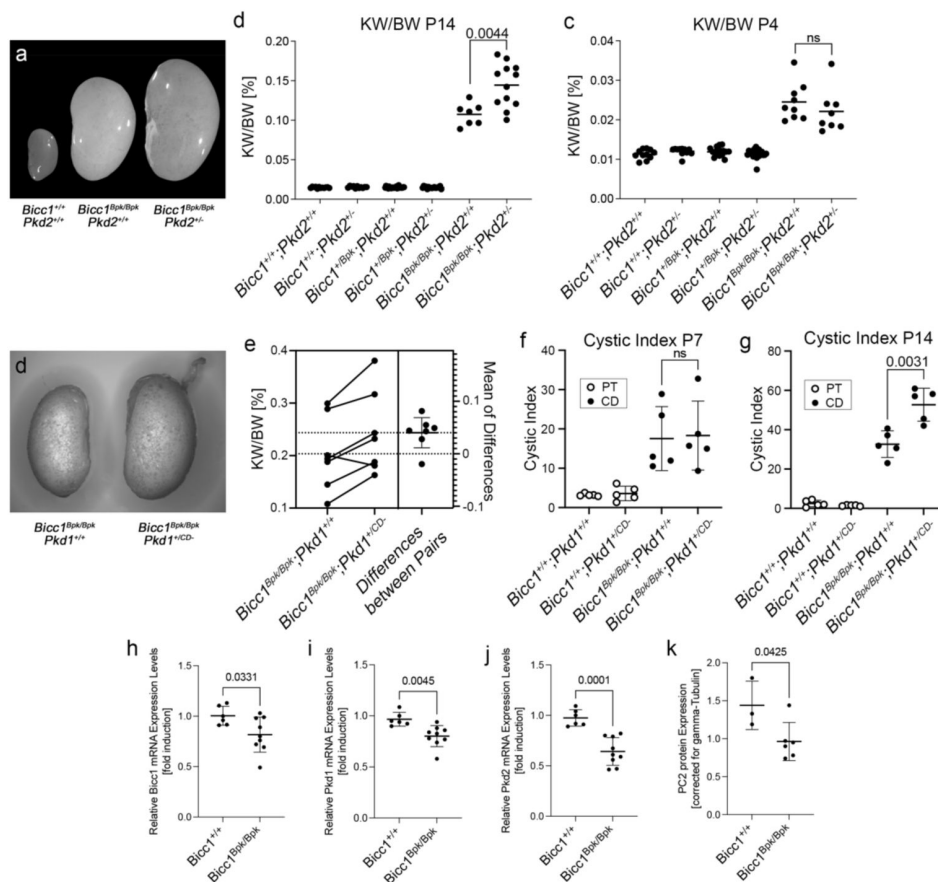


Figure 4

Cooperativity of Bicc1 and Pkd1 and Pkd2 in Mouse.

(a-c) Bicc1 and Pkd2 interact genetically. Offspring from *Bicc1*;*Pkd2* compound mice at postnatal day P4 and P14 are compared by outside kidney morphology at postnatal day P14 (a), and kidney to body weight ratio (KW/BW) at P14 (b) and P4 (c). (d-g) Bicc1 and Pkd1 interact genetically. *Bicc1*;*Pkd1* compound mice are compared by outside kidney morphology at P14 (d), estimation plot of KW/BW ratio comparing littermates at P14 with a *P*-value = 0.092 (e), and cystic index of proximal tubules (PT) and collecting ducts (CD) at P7 (f) and P14 (g). Two-sided paired t-tests were performed to assess significance and the *P*-values are indicated; error bars represent standard deviation. (h-k) qRT-PCR analysis for *Bicc1*, *Pkd1*, and *Pkd2* expression (h-j) and quantification of the PC2 expression levels by Western blot (k) in kidneys at P4 before the onset of a strong cystic kidney phenotype. Data were analyzed by t-test and the *P*-values are indicated.

(SIFT, Polyphen2, CADD, Eigen-PC, FATHMM, GERP++ RS and EVE), meta scores (REVEL, MetaSVM, and MetaLR) and other protein function predictions (PrimateAI, Alphasense, ESM1b and ProtVar) indicate that this PKD2 missense variant is likely pathogenic (**Supplementary Table S3**). Moreover, structural analysis suggests that the hydrophilic substitution may interfere with the Helix S5 pore domain of PKD2 and change its ion channel function (**Fig. 5b,c**). Finally, PKD2 p.Cys632Arg has been previously reported as part of a PKD2 pedigree and implicated as a critical determinant for Polycystin-2 function.^{43, 44} On the other hand, the BICC1 p.Gly821Glu variant is located in an intrinsically disordered domain of BICC1 between the KH and the SAM domains (**Fig. 6f**). To address whether the variant is hypomorphic, we used CRISPR-Cas9-mediated gene editing to generate HEK293T cells lacking BICC1 or harboring the G821E mutation. These cells were analyzed for their impact on the translation of *PKD2*, a well-established target of Bicc1.²² As shown in **Fig. 5d,e** PC2 protein levels were strongly reduced in two independent HEK293T BICC1 (G821E) cells when compared to unedited HEK293T cells. Most notably, the PC2 levels were comparable to the levels found in HEK293T carrying a BICC1 null allele (HEK293T BICC1 KO) (**Supplementary Figure S2c,d**). Based on these data we hypothesize that the major disease effect results from the pathogenic *PKD2* variant but is aggravated by the BICC1 variant.

The second patient presented perinatally with massively enlarged hyperechogenic kidneys, while the parents, both in their thirties, and the remaining family members were reported to be healthy (**Fig. 5f-h**). He carried a paternal canonic *BICC1* pathogenic splicing variant (c.1179+1G>T), which is likely pathogenic as the protein is truncated after exon 10, and a novel heterozygous *PKD1* variant (c.11942C>T, p.Ala3981Val) which has not been previously reported (**Fig. 5f**). While the Polycystin-1 variant appears minor in its amino acid change (i.e. Ala to Val), *in silico* analyses using individual predictions (SIFT, Polyphen2, CADD and EVE), Meta scores (REVEL) and other protein function predictions (PrimateAI, Alphasense and ESM1b) indicate that missense variant is likely pathogenic (**Supplementary Table S3**). Structural analyses suggest that the Ala3981Val variant may not destabilize the Helix structure but may contact the TOP domain and interfere with domain flexibility and PC1 complex assembly.

A Sibling Pair of PKD Patients with a Homozygous *BICC1* Mutation

The most insightful finding for a critical role for BICC1 in human PKD was the discovery of a homozygous *BICC1* variant in a consanguineous Arab multiplex pedigree, two siblings, a boy and a girl, diagnosed with VEO-ADPKD (**Fig. 6a-e**). The affected female presented at a few months of age with kidney failure and enlarged polycystic kidneys that lacked corticomedullary differentiation. Histology after bilateral nephrectomy showed polycystic kidneys more suggestive of ADPKD than ARPKD without any dysplastic element (**Fig. 6c**). Her younger brother exhibited enlarged hyperechogenic polycystic kidneys antenatally by ultrasound (**Fig. 6b**). In addition, during early infancy, arterial hypertension and a Dandy-Walker malformation with a low-pressure communicating hydrocephalus were noted (**Fig. 6d,e**). By customized MPS, we identified the homozygous evolutionarily well-conserved missense variant (c.718T>C, p.Ser240Pro) in *BICC1* (**Fig. 6a**). This variant was absent from gnomAD and perfectly segregated with the cystic phenotype present in this family. It results in a non-conservative change from the aliphatic, polar-hydrophilic serine to the cyclic, apolar-hydrophobic proline located in the second beta sheet of the first KHL1 domain and very likely disrupts the beta sheet and thus the RNA-binding activity of Bicc1 (**Fig. 6f,g** and **Supplementary Table S4**). In the more severely affected younger brother, we also detected the heterozygous *PKD2* change (c.1445T>G, p.Phe482Cys), which results in a non-conservative change from phenylalanine to cysteine (**Supplementary Table S3**). It was previously reported that this Phe482Cys variant exhibited altered kinetic PC2 channel properties, increased expression in IMCD cells and a different subcellular distribution when compared to wild-type PC2.⁴⁵ These features suggested altered properties of this PC2 variant, yet its contribution to the case reported here remain untested.

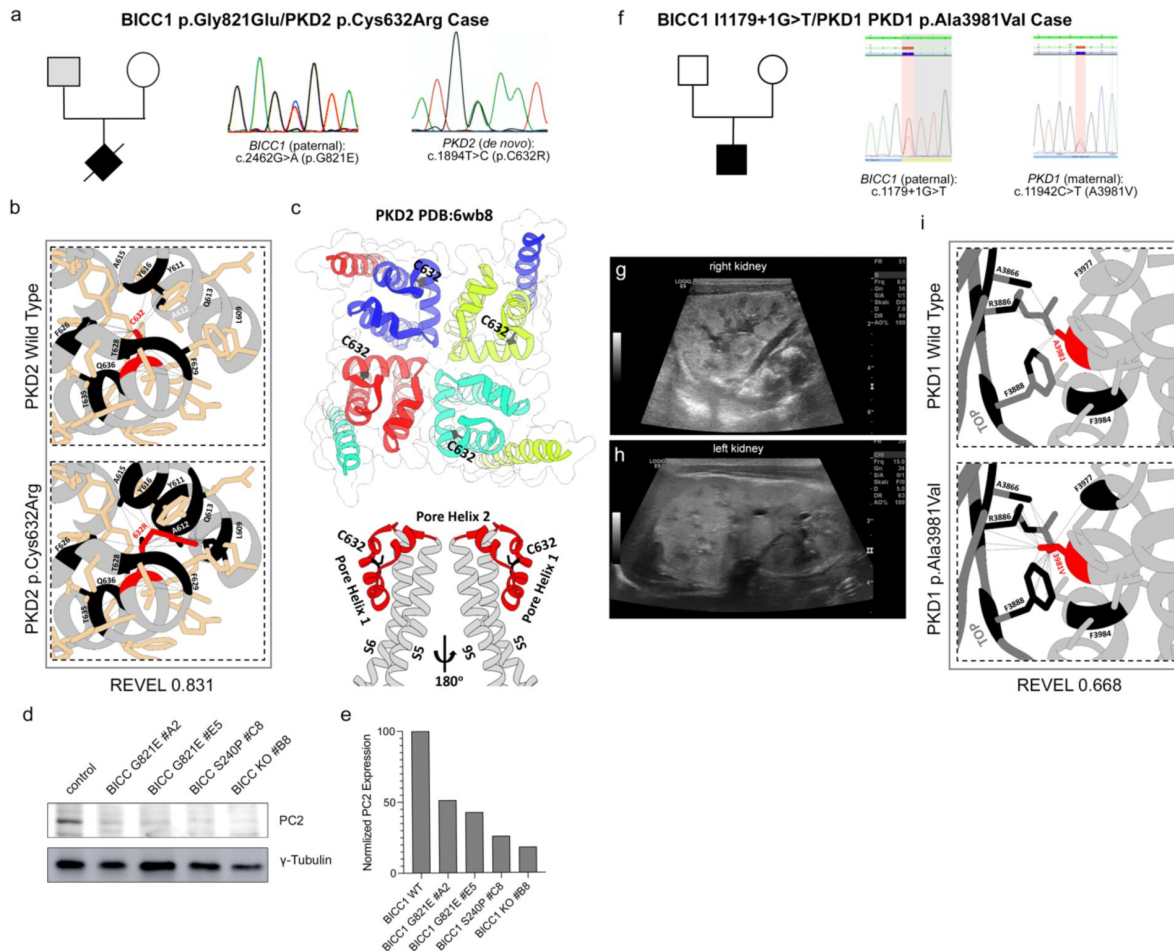


Figure 5.

Identification of Human *BICC1* Variants.

(a-c) BICC1 p.G821E/PKD2 p.C632R case with pedigree and the electropherograms (a), the structural analysis of the PKD2 showing the local structure around the cysteine at position 632 (indicated in red) and its putative impact in the variant including the REVEL score (b) as well as its location within the global PC2 structure highlighting the potential of the variant impacting the PC2 ion channel function (c). (d,e) Western blot analysis for PC2 comparing wildtype HEK293T, BICC1 G821E, BICC1 S240P and BICC1 KO HEK293T and quantification thereof. γ-Tubulin was used as loading control. (f-i) BICC1 I1179+1G>T/PKD1 PKD1 p.Ala3981Val case with pedigree and the electropherograms (f), the ultrasound analysis of the left and right kidneys (g,h) and the structural analysis of the PC1 showing the local structure around the alanine at position 3981 (indicated in red) and its putative impact in the variant including the REVEL score (i).

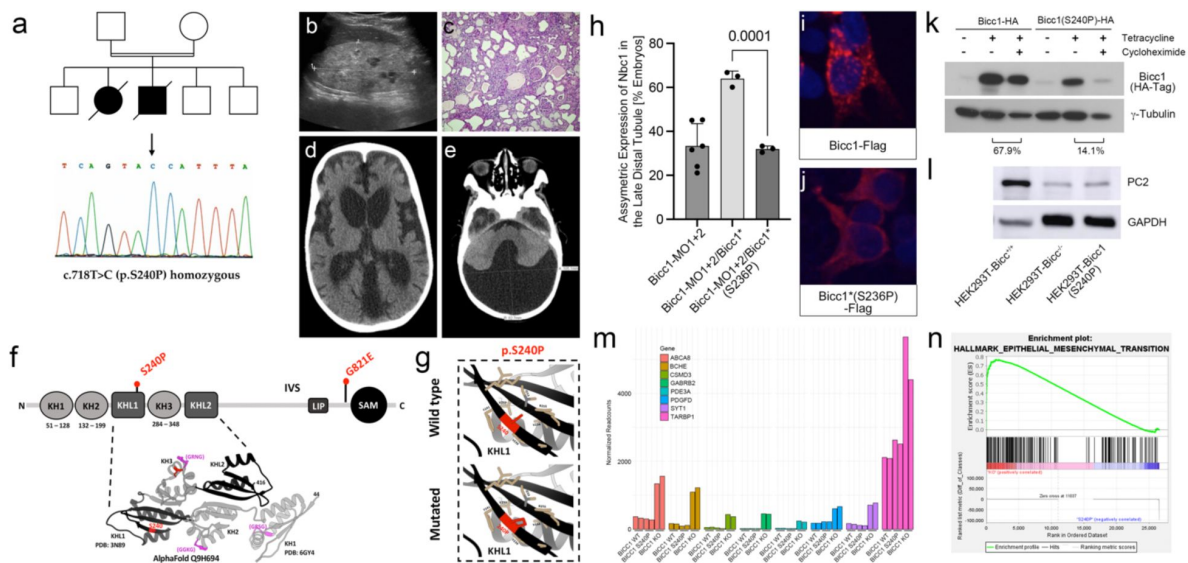


Figure 6

The Homozygous BICC1 S240P Mutation is a Hypomorphic Cystic Disease-Causing Variant.

(a-e) Consanguineous multiplex pedigree with two siblings affected by VEO-ADPKD identified the homozygous *BICC1* missense variant c.718T>C (p.S240P) absent from gnomAD and other internal and public databases. Electropherogram is shown in (a). The affected girl presented at a few months of age with renal failure and enlarged polycystic kidneys that lacked corticomedullary differentiation (c). Histology after bilateral nephrectomy showed polycystic kidneys more suggestive of ADPKD than ARPKD without any dysplastic element. Her younger brother exhibited enlarged hyperechogenic polycystic kidneys prenatally by ultrasound (b). In addition, in his early infancy arterial hypertension and a Dandy-Walker malformation with a low-pressure communicating hydrocephalus were noted (d,e). (f) Ribbon diagram and schematic diagram of BICC1 contains KH, KHL and SAM domains showing the variants discovered. (g) Solid boxes correspond to local impacts of p.S240P on BICC1 structure, interactions are labeled as dashed lines (pseudobonds). GXXG motifs colored in magenta, representative missense variant residues colored in red and residues adjacent to selected variant (<5Å) colored in tan. (h) Rescue experiments of *Xenopus* embryos lacking Bicc1 by co-injections with the wild type or mutant constructs. Embryos were scored for the re-expression of *Nbc1* in the late distal tubule by whole mount *in situ* hybridizations. Quantification of at least 3 independent experiments is shown. (i,j) HEK293T cells were transfected with Flag-tagged constructs of wild type or mutant Bicc1 and the subcellular localization of Bicc1 was visualized (red). Nuclei were counterstained with DAPI (blue). (k) Protein stability analysis using tetracycline-inducible HEK293T cells comparing the expression levels of Bicc1 and Bicc1(S240P) 24 hours after removal of tetracycline and addition of cycloheximide. γ -Tubulin was used as loading control. The percentage of protein destabilization because of protein synthesis inhibition by cycloheximide is indicated. (l) Western Blot analysis of wildtype HEK293T, cells lacking BICC1 (*BICC1*^{-/-}) and isogenic cells with the BICC1 S240P mutation for PC2 expression. GAPDH was used as loading control. (m,n) Bar graph of the mRNA-seq transcriptomic analysis comparing BICC1 wildtype, knockout and S240P isogenic HEK293T cells showing the eight most significantly upregulated transcripts (based on their P_{adj} levels) in the BICC1 KO cells (m). For each gene, the normalized expression levels from each of the 6 samples (2 wildtype, KO and 240P each) is shown. (n) GSEA plot showing the enrichment of the Hallmark Epithelial_Mesenchymal_Transition data set in the BICC1 KO cells vs. the BICC1 S240P cells.

Unfortunately, both siblings passed away and besides DNA and the phenotypic analysis described above neither human tissue nor primary patient-derived cells could be collected. Thus, to validate the pathogenicity of this point mutation, we turned to the amphibian model of PKD.^{21, 22} In *Xenopus*, knockdown of *Bicc1* using antisense morpholino oligomers (*Bicc1-MO1+2*) causes a PKD phenotype, which can be rescued by co-injection of synthetic mRNA encoding *Bicc1*.²¹ To test whether *BICC1* p.(Ser240Pro) has lost its biological activity, we introduced the same mutation into the *Xenopus* gene (*Bicc1** *S236P*). *Xenopus* embryos were injected with *Bicc1-MO1+2* at the 2-4 cell stage followed by a single injection of 2 ng wild type or *Bicc1** *S236P* mRNAs at the 8-cell stage. At stage 39 (when kidney development has been completed) embryos were analyzed by whole mount *in situ* hybridization for the expression of *Nbc1* in the late distal tubule of the pronephric kidney, one of the most reliable readouts for the amphibian PKD phenotype.²¹ As shown in **Fig. 6h**, wild type *Bicc1* mRNA restored expression of *Nbc1* on the injected side in 63% of the embryos. However, *Bicc1** *S236P* did not have any effect, and the embryos were indistinguishable from those injected with the *Bicc1-MO1+2* alone. This suggested that *Bicc1** *S236P* was functionally impaired. To address this hypothesis, we first assessed the subcellular localization of *Bicc1* to foci that are thought to be involved in mRNA processing.^{19, 22, 27, 46} Transfection of Flag-tagged *Bicc1* into HEK293T cells reproduced this pattern (**Fig. 6i**). Surprisingly, *Bicc1* *S236P*-Flag was no longer detected in these cytoplasmic foci but rather homogenously dispersed throughout the cytoplasm (**Fig. 6j**). Western blot analysis demonstrated that this was accompanied by a reduction in protein levels (**Fig. 6k**). *In vitro* transcription/translation detected no differences between the proteins suggesting that the wildtype and *Bicc1* *S236P*-Flag are translated equivalently (data not shown). Yet, in an *in vivo* pulse-chase experiments the *S240P* variant was less stable than its wildtype counterpart (**Fig. 6k**). However, whether the reduced protein level was due to an inherent instability of the mutant protein or a consequence of its mislocalization remains to be resolved. Finally, as in the case of *BICC1* G821E we engineered HEK293T cells to harbor the *BICC1* *S240P* mutation. Western blot analysis demonstrated a reduction in PC2 levels in the *BICC1* *S240P* cells when compared to unedited cells and that this reduction was comparable to PC2 levels in *BICC1* KO cells (**Figs. 5d,e** and **6l**).

Finally, to determine to what extent the *BICC1* *S240P* variant differs from a *BICC1* loss of function, we performed mRNA sequencing (mRNA-seq) of the genetically engineered HEK293T cells. Differential gene expression analysis identified several genes that were differentially up-or down-regulated in the *BICC1* *S240P* and the *BICC1* KO cells compared to their unedited counterpart (**Supplementary Fig. 4a,e**). Approximately 24% and 18% of the differentially expressed genes were shared between *BICC1* *S240P* or the KO cells, respectively (**Supplementary Fig. S4b,f**). Yet, a substantial number of genes were specific to either cell line. The *BICC1* *S240P*-enriched/depleted transcripts were generally also enriched/depleted in the *BICC1* KO cells, but did not reach statistical significance (**Supplementary Fig. 4c,g**). Conversely, many of the *BICC1* KO enriched transcripts were specifically enriched/depleted in the *BICC1* KO cells and not in the *BICC1* *S240P* cells (**Fig. 6m** and **Supplementary Fig. 4d**). This suggested that there are qualitative differences between a null phenotype and the *BICC1* *S240P* variant, supporting our hypothesis that *BICC1* *S240P* acts as a hypomorph. Indeed, Gene Set Enrichment Analysis (GSEA) using the hallmark gene sets and comparing *BICC1* KO and *S240P* cells revealed a statistically significant enrichment for the Hallmark_Epithelial_Mesenchymal_Transition set (**Fig. 6n** and **Supplementary Table S5**), a pathway previously implicated in ADPKD.^{47, 48}

Discussion

BICC1 has been extensively studied in multiple animal models, which have suggested a critical role for *BICC1* in several different developmental processes and in tissue homeostasis.²³ This study functionally implicates it to human disease in general and PKD in particular by identifying the homozygous *BICC1* *S240P* variant, which was sufficient to cause a cystic phenotype in a sibling pair of human PKD patients. It is noteworthy that another study identified heterozygous *BICC1*

mutations in two patients with mildly cystic dysplastic kidneys.⁴⁹ Yet, the variants were both also present in one of the parent. While such a situation is extremely rare and does not significantly contribute to the mutational load in ADPKD or ARPKD, it demonstrated that loss of *BICC1* is sufficient to cause PKD in humans. In addition, variants in *BICC1* and *PKD1* and *PKD2* co-segregated in PKD patients from an International Clinical Diagnostic Cohort. While we have not yet shown the pathogenesis of the variants when introduced as a compound situation, we postulate that PKD alleles *in trans* and/or *de novo* exert an aggravating effect and contribute to polycystic kidney disease. A reduced dosage of PKD proteins would severely disturb the homeostasis and network integrity, and by this correlates with disease severity in PKD. ADPKD is quite heterogeneous and - even within the same family - shows quite some phenotypic variation.^{50, 51} It is thought that stochastic inputs, environmental factors and genetics influence PKD.⁵¹ The demonstrated interaction of *BICC1*, *PC1* and *PC2* now provides a molecular mechanism that can explain some of the phenotypic variability in these families. Of note, while our mouse studies support the cooperation between *Bicc1*, *Pkd1* and *Pkd2*, the genetic proof for *Bicc1* acting as a disease modifier, i.e. reduction of *Bicc1* activity in a heterozygous *Pkd1* or *Pkd2* background is still outstanding.

The second important aspect of the study is that *BICC1* emerges as a central in the regulation of *PKD1*/*PKD2* activity. Functional studies reported here and previously^{22, 32, 33} demonstrate that *BICC1* regulates the expression of *PKD1* and *PKD2*. Moreover, we now show that *BICC1* and *PC1*/*PC2* physically interact and that lowering the expression levels of both proteins was sufficient to cause a PKD phenotype in frogs. Finally, in mice the reduction of the gene dose for *Pkd1* or *Pkd2* in a hypomorphic allele of *Bicc1* results in a more severe cystic kidney phenotype. These results in the kidney are paralleled and augmented in studies of left/right patterning, where *PC2* can activate *Bicc1* and where *Bicc1* triggers critical aspects in establishing laterality.^{19, 27, 52, 53} Thus, it is tempting to speculate that *BICC1*/*PKD1*/*PKD2* are components of a critical regulatory network in maintaining epithelial homeostasis. *BICC1* functions as a posttranscriptional regulator modifying gene expression modulating the effects of microRNAs (miRNAs), regulating mRNA polyadenylation and translational repression and activation.^{22, 23, 29, 54–57} While *PKD2* is the most appealing target in respect to ADPKD, there are undoubtable others (e.g. adenylate cyclase-6)²⁹ that are equally critical. In the future, it will be interesting to see how all these targets are integrated and together contribute towards protecting kidney epithelial cells from undergoing cystic malformations. Lastly, *BICC1* has been implicated in the regulation of miRNAs such as the ones of the *miR-17* family.²² A connection between *miR-17* activity and PKD is well-established^{58–63} and is even considered as a PKD therapeutic.⁶⁴ Thus, the interaction between *BICC1*, *PKD1* and *PKD2* and miRNAs - even though not examined in this study - may be the ultimate regulatory complex that is responsible for several of the aspects of human ADPKD.

Data sharing statement

The datasets are presented in the figures and the supplementary information. The mRNA-seq data are deposited into the Gene Expression Omnibus (GEO) database (GSE262417) and are available online. Additional information is available from the corresponding author on reasonable request.

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Additional files

Supplemental Material [↗](#)

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Reviewer #1 (Public review):

In this manuscript, Tran et al. investigate the interaction between BICC1 and ADPKD genes in renal cystogenesis. Using biochemical approaches, they reveal a physical association between

Bicc1 and PC1 or PC2 and identify the motifs in each protein required for binding. Through genetic analyses, they demonstrate that Bicc1 inactivation synergizes with Pkd1 or Pkd2 inactivation to exacerbate PKD-associated phenotypes in *Xenopus* embryos and potentially in mouse models. Furthermore, by analyzing a large cohort of PKD patients, the authors identify compound BICC1 variants alongside PKD1 or PKD2 variants in trans, as well as homozygous BICC1 variants in patients with early-onset and severe disease presentation. They also show that these BICC1 variants repress PC2 expression in cultured cells.

Overall, the concept that BICC1 variants modify PKD severity is plausible, the data are robust, and the conclusions are largely supported. However, several aspects of the study require clarification and discussion:

- (1) The authors devote significant effort to characterizing the physical interaction between Bicc1 and Pkd2. However, the study does not examine or discuss how this interaction relates to Bicc1's well-established role in posttranscriptional regulation of Pkd2 mRNA stability and translation efficiency.
- (2) Bicc1 inactivation appears to downregulate Pkd1 expression, yet it remains unclear whether Bicc1 regulates Pkd1 through direct interaction or by antagonizing miR-17, as observed in Pkd2 regulation. This should be further examined or discussed.
- (3) The evidence supporting Bicc1 and ADPKD gene cooperativity, particularly with Pkd1, in mouse models is not entirely convincing, likely due to substantial variability and the aggressive nature of Bpk/Bpk mice. Increasing the number of animals or using a milder Bicc1 strain, such as jcpk heterozygotes, could help substantiate the genetic interaction.

<https://doi.org/10.7554/eLife.106342.1.sa3>

Reviewer #2 (Public review):

Tran and colleagues report evidence supporting the expected yet undemonstrated interaction between the Pkd1 and Pkd2 gene products Pc1 and Pc2 and the Bicc1 protein in vitro, in mice, and collaterally, in *Xenopus* and HEK293T cells. The authors go on to convincingly identify two large and non-overlapping regions of the Bicc1 protein important for each interaction and to perform gene dosage experiments in mice that suggest that Bicc1 loss of function may compound with Pkd1 and Pkd2 decreased function, resulting in PKD-like renal phenotypes of different severity. These results led to examining a cohort of very early onset PKD patients to find three instances of co-existing mutations in PKD1 (or PKD2) and BICC1. Finally, preliminary transcriptomics of edited lines gave variable and subtle differences that align with the theme that Bicc1 may contribute to the PKD defects, yet are mechanistically inconclusive.

These results are potentially interesting, despite the limitation, also recognized by the authors, that BICC1 mutations seem exceedingly rare in PKD patients and may not "significantly contribute to the mutational load in ADPKD or ARPKD". The manuscript has several intrinsic limitations that must be addressed.

The manuscript contains factual errors, imprecisions, and language ambiguities. This has the effect of making this reviewer wonder how thorough the research reported and analyses have been.

<https://doi.org/10.7554/eLife.106342.1.sa2>

Reviewer #3 (Public review):

Summary:

This study investigates the role of BICC1 in the regulation of PKD1 and PKD2 and its impact on cystogenesis in ADPKD. By utilizing co-IP and functional assays, the authors demonstrate physical, functional, and regulatory interactions between these three proteins.

Strengths:

- (1) The scientific principles and methodology adopted in this study are excellent, logical, and reveal important insights into the molecular basis of cystogenesis.
- (2) The functional studies in animal models provide tantalizing data that may lead to a further understanding and may consequently lead to the ultimate goal of finding a molecular therapy for this incurable condition.
- (3) In describing the patients from the Arab cohort, the authors have provided excellent human data for further investigation in large ADPKD cohorts. Even though there was no patient material available, such as HUREC, the authors have studied the effects of BICC1 mutations and demonstrated its functional importance in a *Xenopus* model.

Weaknesses:

This is a well-conducted study and could have been even more impactful if primary patient material was available to the authors. A further study in HUREC cells investigating the critical regulatory role of BICC1 and potential interaction with mir-17 may yet lead to a modifiable therapeutic target.

Conclusion:

The authors achieve their aims. The results reliably demonstrate the physical and functional interaction between BICC1 and PKD1/PKD2 genes and their products.

The impact is hopefully going to be manifold:

- (1) Progressing the understanding of the regulation of the expression of PKD1/PKD2 genes.
- (2) Role of BICC1 in mir/PKD1/2 complex should be the next step in the quest for a modifiable therapeutic target.

<https://doi.org/10.7554/eLife.106342.1.sa1>

Author response:**Reviewer #1 (Public Review):**

*In this manuscript, Tran et al. investigate the interaction between BICC1 and ADPKD genes in renal cystogenesis. Using biochemical approaches, they reveal a physical association between Bicc1 and PC1 or PC2 and identify the motifs in each protein required for binding. Through genetic analyses, they demonstrate that Bicc1 inactivation synergizes with Pkd1 or Pkd2 inactivation to exacerbate PKD-associated phenotypes in *Xenopus* embryos and potentially in mouse models. Furthermore, by analyzing a large cohort of PKD patients, the authors identify compound BICC1 variants alongside PKD1 or PKD2 variants in trans, as well as homozygous BICC1 variants in patients with early-onset and severe disease presentation. They also show that these BICC1 variants repress PC2 expression in cultured cells.*

Overall, the concept that BICC1 variants modify PKD severity is plausible, the data are robust, and the conclusions are largely supported. However, several aspects of the study require clarification and discussion:

(1) The authors devote significant effort to characterizing the physical interaction between Bicc1 and Pkd2. However, the study does not examine or discuss how this interaction relates to Bicc1's well-established role in posttranscriptional regulation of Pkd2 mRNA stability and translation efficiency.

The reviewer is correct that the present study has not addressed the downstream consequences of this interaction considering that Bicc1 is a posttranscriptional regulator of Pkd2 (and potentially Pkd1). We think that the complex of Bicc1/Pkd1/Pkd2 retains Bicc1 in the cytoplasm and thus restrict its activity in participating in posttranscriptional regulation. As we do not have yet experimental data to support this model, we have not included this model in the manuscript. Yet, we will update the discussion of the manuscript to further elaborate on the potential mechanism of the Bicc1/Pkd1/Pkd2 complex.

(2) Bicc1 inactivation appears to downregulate Pkd1 expression, yet it remains unclear whether Bicc1 regulates Pkd1 through direct interaction or by antagonizing miR-17, as observed in Pkd2 regulation. This should be further examined or discussed.

This is a very interesting comment. The group of Vishal Patel published that PKD1 is regulated by a *mir-17* binding site in its 3'UTR (PMID: 35965273). We, however, have not evaluated whether BICC1 participates in this regulation. A definitive answer would require us utilize some of the mice described in above reference, which is beyond the scope of this manuscript. We, however, will revise the discussion to elaborate on this potential mechanism.

(3) The evidence supporting Bicc1 and ADPKD gene cooperativity, particularly with Pkd1, in mouse models is not entirely convincing, likely due to substantial variability and the aggressive nature of Bpk/Bpk mice. Increasing the number of animals or using a milder Bicc1 strain, such as jcpk heterozygotes, could help substantiate the genetic interaction.

We have initially performed the analysis using our Bicc1 complete knockout, we previously reported on (PMID 20215348) focusing on compound heterozygotes. Yet, like the Pkd1/Pkd2 compound heterozygotes (PMID 12140187) no cyst development was observed until we sacrificed the mice at P21. Our strain is similar to the above mentioned jcpk, which is characterized by a short, abnormal transcript thought to result in a null allele (PMID: 12682776). We thank the reviewer for pointing use to the reference showing the heterozygous mice show glomerular cysts in the adults (PMID: 7723240). This suggestion is an interesting idea we will investigate. In general, we agree with the reviewer that the better understanding the contribution of Bicc1 to the adult PKD phenotype will be critical. To this end, we are currently generating a floxed allele of Bicc1 that will allow us to address the cooperativity in the adult kidney, when e.g. crossed to the Pkd1^{RC/RC} mice. Yet, these experiments are unfortunately beyond the scope of this manuscript.

Reviewer #2 (Public Review):

Tran and colleagues report evidence supporting the expected yet undemonstrated interaction between the Pkd1 and Pkd2 gene products Pc1 and Pc2 and the Bicc1 protein in vitro, in mice, and collaterally, in Xenopus and HEK293T cells. The authors go on to convincingly identify two large and non-overlapping regions of the Bicc1 protein important for each interaction and to perform gene dosage experiments in mice that suggest that Bicc1 loss of function may compound with Pkd1 and Pkd2 decreased

function, resulting in PKD-like renal phenotypes of different severity. These results led to examining a cohort of very early onset PKD patients to find three instances of co-existing mutations in PKD1 (or PKD2) and BICC1. Finally, preliminary transcriptomics of edited lines gave variable and subtle differences that align with the theme that Bicc1 may contribute to the PKD defects, yet are mechanistically inconclusive.

These results are potentially interesting, despite the limitation, also recognized by the authors, that BICC1 mutations seem exceedingly rare in PKD patients and may not "significantly contribute to the mutational load in ADPKD or ARPKD". The manuscript has several intrinsic limitations that must be addressed.

As mentioned above, the study was designed to explore whether there is an interaction between BICC1 and the PKD1/PKD2 and whether this interaction is functionally important. How this translates into the clinical relevance will require additional studies (and we have addressed this in the discussion of the manuscript).

The manuscript contains factual errors, imprecisions, and language ambiguities. This has the effect of making this reviewer wonder how thorough the research reported and analyses have been.

We respectfully disagree with the reviewer on the latter interpretation. The study was performed with rigor. We have carefully assessed the critiques raised by the reviewer. Most of the criticisms raised by the reviewer will be easily addressed in the revised version of the manuscript. Yet, none of the critiques raised by the reviewer seems to directly impact the overall interpretation of the data.

Reviewer #3 (Public Review):

Summary:

This study investigates the role of BICC1 in the regulation of PKD1 and PKD2 and its impact on cytogenesis in ADPKD. By utilizing co-IP and functional assays, the authors demonstrate physical, functional, and regulatory interactions between these three proteins.

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Weaknesses:

This is a well-conducted study and could have been even more impactful if primary patient material was available to the authors. A further study in HUREC cells investigating the critical regulatory role of BICC1 and potential interaction with mir-17 may yet lead to a modifiable therapeutic target.

This is an excellent suggestion. We agree with the reviewer that it would have been interesting to analyze HUREC material from the affected patients. Unfortunately, besides DNA and the phenotypic analysis described in the manuscript neither human tissue nor primary patient-derived cells collected before the two patients with the BICC1 p.Ser240Pro mutation passed away. To address this missing link, we have – as a first pass - generated HEK293T cells carrying the BICC1 p.Ser240Pro variant. While these admittedly are not kidney epithelial cells, they indeed show a reduced level of PC2 expression. These data are shown in the manuscript. We have not yet addressed how this relates to its crosstalk with *miR-17*.

Conclusion:

The authors achieve their aims. The results reliably demonstrate the physical and functional interaction between BICC1 and PKD1/PKD2 genes and their products.

The impact is hopefully going to be manifold:

(1) Progressing the understanding of the regulation of the expression of PKD1/PKD2 genes.

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